



Management Information Systems (MIS)

Business Process Identification Worksheet

Course: Management Information Systems

Objective: Guide students in identifying business processes to develop an MIS that delivers tangible value to business managers using the Design Thinking methodology.

Team Information

Please fill out your team members' details below:

Role	Full Name	Student ID
Team Leader	Aldi	24523023
Member 1	Abdullah	24523229
Member 2	Kemal	24523123
Member 3	Dani	24523207

Part 1: Business Area Selection

Review the following business areas and select **ONE** for your team's project.

Available Business Areas:

1. **Education:** e.g., Final Project/Thesis Management System (automated thesis title registration and advisor recommendations to balance lecturer workloads).
2. **Healthcare:** e.g., Information System for primary healthcare facilities/clinics (patient registration, early detection, monthly disease trends).

3. **Retail/MSMEs & Logistics Systems:** Micro, Small, and Medium Enterprises, inventory tracking, and supply chain logistics.
4. **Organizational HR:** Human Resources management (job grading, performance appraisal, attendance, etc.).
5. **Food & Beverage (F&B):** From sales recording/POS to menu profitability and ingredient analysis.
6. **Agribusiness:** Farm management, crop tracking, supply chain, and yield prediction.
7. **Tourism:** Booking systems, destination management, visitor tracking, and tour packages.
8. **Public Services:** Community services (Perspectives can range from the hamlet/village level up to the sub-district/district level).
9. **Event Organizing (EO):** Event planning, ticketing, vendor management, and financial reporting.

Chosen Business Area	Specific Context/Company Concept (Optional)
<i>Public Services</i>	Community Resource Management System for BUMDes: A system designed to help village management allocate local human resources to village projects, monitor participation, and track revenue to support decision-making.

Part 2: Design Thinking Worksheet

Work with your team to fill in the tabulated sections below. Ensure you are focusing on how the processes will benefit the **manager**.

Step 1: Empathize

Objective: Understand the manager's perspective, their challenges, and what value the business area delivers.

Guiding Questions	Team's Analysis & Findings
Who are the key managers or stakeholders in this area?	1. BUMDes Director (main manager) 2. Village Head (Lurah) 3. RT/RW (support role, data collection) 4. Residents (workforce/providers)

Guiding Questions	Team's Analysis & Findings
What are their daily operational challenges and pain points?	1. Difficulty assigning local workers to village projects efficiently 2. Lack of structured data on residents' skills and capabilities 3. No system to track worker participation, performance, and contributions 4. Poor visibility of project progress and outcomes 5. Lack of financial tracking related to project revenue and resource usage
What specific value does this business area create for its customers and stakeholders?	1. Improves efficiency in executing village economic projects 2. Enables better utilization of local human resources 3. Ensures fair distribution of job opportunities to residents 4. Supports transparent and data-driven decision-making 5. Contributes to revenue generation and local economic development

Step 2: Define

Objective: State the core problems clearly and define the specific value the MIS needs to deliver to the manager.

Guiding Questions	Team's Analysis & Findings
What is the primary problem or bottleneck we are trying to solve?	Lack of a structured system to manage and allocate local human resources to village economic activities, resulting in inefficient project execution, missed revenue opportunities, and poor decision-making.
What specific management benefits/values must our MIS provide to the manager?	1. Real-time visibility of available workforce (skills and availability) 2. Efficient workforce allocation to village projects

Guiding Questions	Team's Analysis & Findings
	3. Ability to monitor project progress and worker performance 4. Revenue tracking and financial insights 5. Reports and analytics to support planning and decision-making

Step 3: Ideate

Objective: Brainstorm what business processes are required to deliver the defined value.

Guiding Questions	Team's Analysis & Findings
What are the potential business processes needed? (List at least 3-5 ideas)	1. Workforce Skills Registration Process (Residents register their skills, experience, and availability) 2. Project Creation & Management Process (Manager creates and defines village economic projects) 3. Worker Assignment Process (System matches and assigns workers based on required skills) 4. Task Execution & Performance Tracking Process (Track work progress, participation, and completion) 5. Reporting & Analytics Process (Generate insights for management decision-making)
Out of these ideas, which business processes are the most critical to support managerial decision-making?	1. Workforce Allocation & Assignment 2. Project Performance Monitoring 3. Revenue & Outcome Reporting

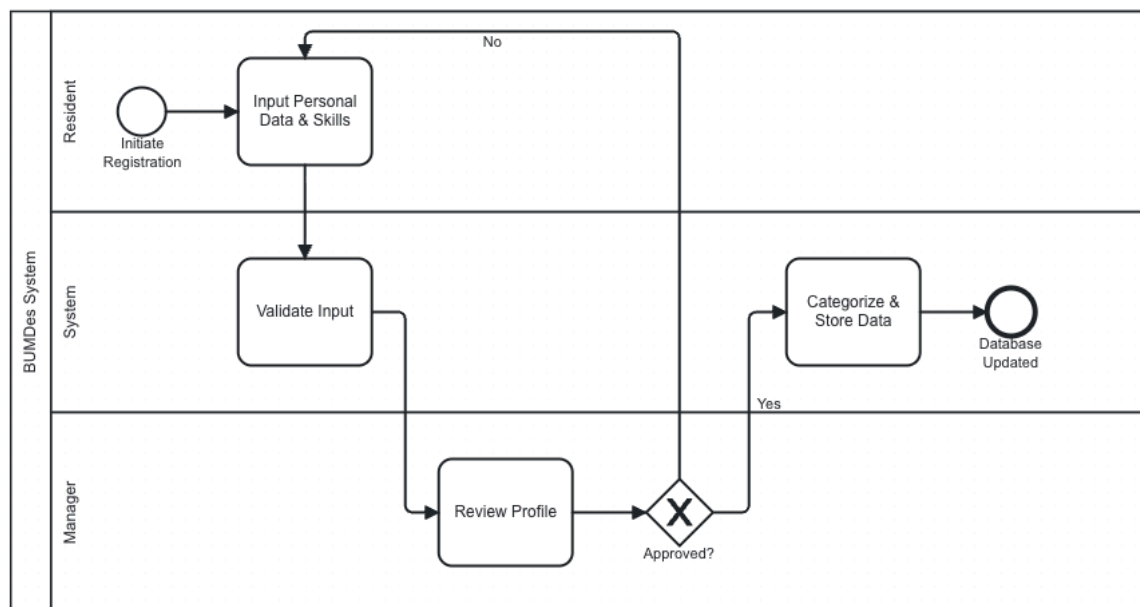
Step 4: Prototype (Business Process Design & BPMN)

Objective: Produce the structured business processes based on your ideation. Map out the step-by-step flow.

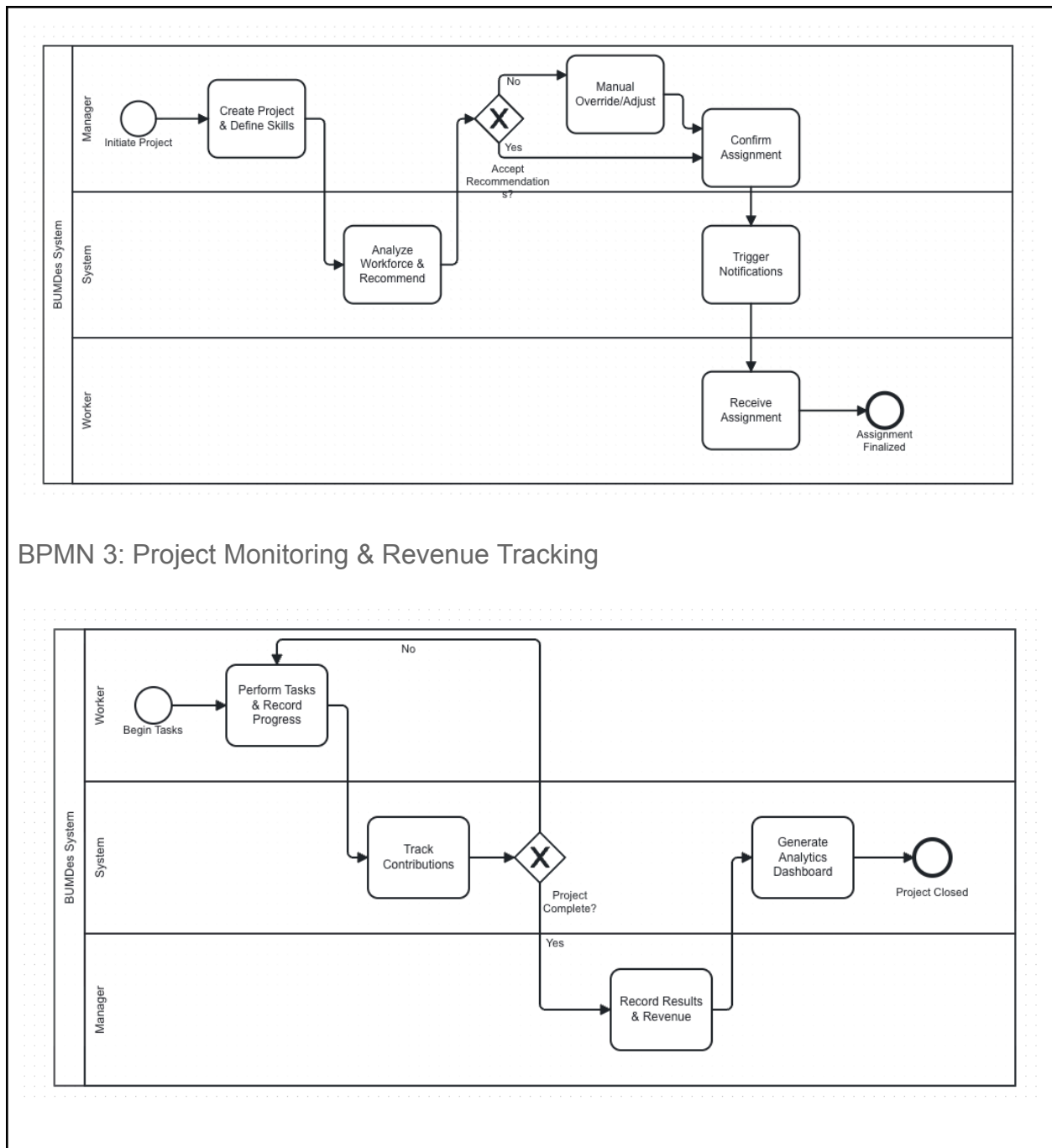
Core Business Process Name	Step-by-Step Flow / Activities	Input Data & Expected Output
Process 1: <i>Workforce Skills Registration</i>	1. Resident creates a profile 2. Resident inputs skills, experience, and availability 3. System categorizes and stores the data 4. Manager reviews and approves (optional)	Input: 1. Resident personal data 2. Skill and experience information Output: Workforce database (structured list of available skills and workers)
Process 2: <i>Project Assignment & Workforce Allocation</i>	1. Manager creates a village project (e.g., farming, tourism, construction) 2. Manager defines required skills and number of workers 3. System analyzes available workforce 4. System recommends suitable workers 5. Manager confirms and assigns workers 6. Workers receive assignment notification	Input: 1. Project details 2. Skill requirements 3. Workforce database Output: 1. Assigned workforce list 2. Project assignment record
Process 3: <i>Project Monitoring & Revenue Tracking</i>	1. Workers perform assigned tasks 2. Work progress is recorded in the system 3. System tracks individual	Input: 1. Task completion data 2. Work progress updates 3. Revenue data

Core Business Process Name	Step-by-Step Flow / Activities	Input Data & Expected Output
	contributions and participation - Project results and revenue are recorded - Dashboard generates performance and financial reports	Output: - Project performance report - Worker contribution report - Revenue summary and analytics

BPMN 1: Workforce Skills Registration



BPMN 2: Project Assignment & Workforce Allocation



Step 5: Test (Manual Simulation)

Objective: Perform a manual simulation (role-play) of the business processes within your team to identify gaps, bottlenecks, or missing steps.

Simulation Phase	Observations & Results	Areas for Improvement
Process 1 Simulation: (Workforce Registration) Resident: Abdullah Manager: Aldi	1. System successfully stores and organizes skill data 2. Some skill entries are inconsistent or unclear	1. Add predefined skill categories 2. Improve input validation
Process 2 Simulation: (Project Assignment) Manager: Aldi Workers: Kemal, Abdullah	1. System can match workers to project requirements 2. Some assignments require manual adjustment	1. Improve matching criteria 2. Allow flexible filtering and manual override
Process 3 Simulation: (Monitoring & Reporting) Workers: Kemal Manager: Dani	1. System provides useful visibility on progress and contributions 2. Reports help evaluate workforce performance	1. Add more detailed analytics (e.g., productivity comparison) 2. Improve dashboard clarity and visualization
Final Conclusion:	<i>Are the mapped processes ready for MIS software development? If not, what needs to change?</i>	The mapped processes are ready for MIS software development. However, improvements are needed in data standardization, matching accuracy, and reporting features to ensure effective workforce management, project execution, and decision support for village management.